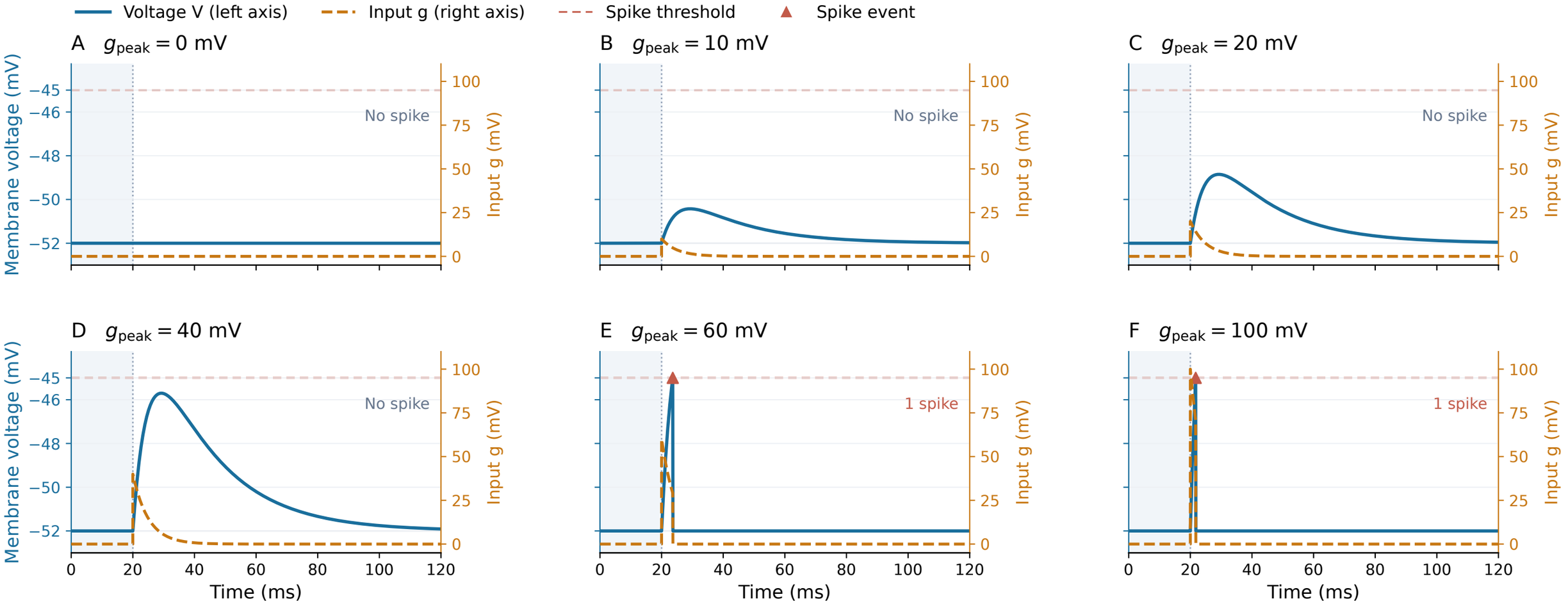
Membrane voltage and synaptic input over time

Single event at 20 ms: $g \leftarrow g + g_{\text{peak}}$, then $\tau_s \, dg/dt = -g$

$\tau_m \, dV/dt = V_{\text{rest}} - V + g$; $\tau_m = 20$ ms, $\tau_s = 5$ ms, $V_{\text{rest}} = V_{\text{reset}} = -52$ mV



All panels use the same axis scales. Input g decays with a 5 ms time constant and resets to 0 at each spike.

Spike threshold: −45 mV; voltage reset: −52 mV; refractory period: 2.2 ms. Action-potential waveforms are not modeled.